



bảng tính

Kiến trúc máy tính PTIT

Tổng cộng questions: 215

Thời gian làm bài: 2 giờ 55 phút

Tên

Lớp học

Ngày

1. Network attached storage is ... which is dedicated for ...
 - a) a machine / storage
 - b) a server / storage
 - c) a server / storing information
 - d) a machine / storing information
2. What are clock speeds of ISA buses
 - a) 6MHz, 8MHz, 10MHz
 - b) 8MHz, 10MHz, 12MHz, 14MHz
 - c) 4MHz, 6MHz, 8MHz
 - d) 2MHz, 4MHz, 6MHz
3. The computer memory system is usually organized in hierarchical levels and the memory hierarchical roles are ...
 - a) Improve the system efficiency and reduce the manufacturing costs
 - b) Improve the system efficiency and reduce the sale costs
 - c) Improve the system performance and reduce the manufacturing costs
 - d) Improve the system performance and reduce the sale costs
4. Common control and sequencing instructions include...
 - a) JUMP, CALL, RETURN, BRANCH-IF-CONDITION
 - b) JMP, CALL, RET, BRANCH-IF-CONDITION
 - c) JUMP, CALL, RET, BRANCH-CONDITION
 - d) JUMP, CALL, RETURN, BRANCH-IF
5. A nibble is a group of ... bits
 - a) 4
 - b) 2
 - c) 6
 - d) 8

6. What is an input peripheral device for computers
- a) Monitor
 - b) CD-RW drive
 - c) Printer
 - d) Plotter
7. Two types of cache architecture are ...
- a) look inside and look through
 - b) look aside and look through
 - c) look aside and look thorough
 - d) look inside and look thorough
8. Each bit of FR register stores the ... of the operation executed by ALU
- a) status of the result
 - b) result
 - c) status
 - d) state of the result
9. What are the purposes in designing RAID storage system
- a) High speed, High durability, Large size
 - b) High performance, High reliability, Large size
 - c) High performance, High reliability, Large volume
 - d) High speed, High durability, Large volume
10. Given instructions: "ADD R2, R3, R4 / CMP R1, 0 / JNE somewhere" and each instruction is executed in 2 stages of IF and EX, how many NO-OP instructions are needed to be inserted after the instruction "JNE somewhere" to solve the branching conflict
- a) 3
 - b) 2
 - c) 1
 - d) 0
11. Which is a true statement about the position of cache in memory system
- a) Cache usually resides inside the CPU
 - b) Cache usually resides outside the CPU
 - c) Cache must reside inside the CPU
 - d) Cache can be located any where in a computer system
12. 0 address of operands is used in instructions that perform ... operations, such as push and pop
- a) tree
 - b) stack
 - c) queue
 - d) string

13. PCI bus uses parallel communication type. What is the communication type of PCIe bus
- a) Point to multi-point
 - b) Serial
 - c) Integrated
 - d) Parallel
14. What are external storages of computer systems
- a) FDD, HDD, SSD, DVD, RAID, NAS, SAN
 - b) FFD, HDD, SSD, DVD, RAD, NAT, SAN
 - c) FDD, HSD, SDD, DVD, RAD, NAS, SAN
 - d) FFD, HDD, SSD, DVD, RAID, NAS, LAN
15. What is an output device
- a) Keyboard
 - b) Scanner
 - c) Mouse
 - d) Disk drive
16. IR or Instruction register is used to store the ... instruction
- a) current executed
 - b) next
 - c) next executed
 - d) next executed
17. Which is NOT a cache organization technique
- a) Set associative mapping
 - b) Fully associative mapping
 - c) Indirect mapping
 - d) Direct mapping
18. What is the communication speed of USB 3.0 ports
- a) 1Gb/s
 - b) 1.5Gb/s
 - c) 2.0Gb/s
 - d) 3Gb/s
19. Internal Bus is the communication channel of all CPU components. It supports ... ways of communication
- a) three
 - b) two
 - c) four
 - d) one

20. What is the correct name of PCI bus
- a) Peripheral Component Interface
 - b) Peripheral Component Interconnect
 - c) Peripher Component Interface
 - d) Peripher Component Interconnect
21. The cache temporal locality usually applies to data items and instructions in a ... of program space
- a) for loop
 - b) while loop
 - c) segment
 - d) loop
22. What are elements of a HDD in computer systems
- a) Disks, heads, racks, sectors, cylinders
 - b) Disk, head, rack, sector, cylinder
 - c) Disks, heads, tracks, sectors, cylinders
 - d) Disks, head, tracks, sectors, cilinders
23. What are types of computer mice
- a) Mechanical mouse, Optical mouse, Laser mouse
 - b) Electronic mouse, Optical mouse, Violet mouse
 - c) Electric mouse, Red light mouse, Laser mouse
 - d) Mechanical mouse, Red light mouse, Laser mouse
24. What are the characteristics of ROM memory
- a) Non-volatile memory, semiconductor memory, read only memory
 - b) Non-volatile memory, semiconductor memory, random access memory
 - c) Volatile memory, magnetic memory, read only memory
 - d) Volatile memory, semiconductor memory, read only memory
25. What is the meaning of the instruction "ADD R3, R2, R1", where Ri is a CPU register
- a) $R3 \leftarrow R3 + R2 + R1$
 - b) $R3 \leftarrow R2 + R1$
 - c) $R1 \leftarrow R2 + R3$
 - d) $R2 \leftarrow R3 + R1$
26. What are the capacity of standard CD-ROM and DVD-ROM
- a) 720MB and 4.8GB
 - b) 750MB and 4.2GB
 - c) 700MB and 4.7GB
 - d) 650MB and 4.5GB

27. What is the correct name of NAS storage system
- a) Net Attacked Storage
 - b) Net Attached Storage
 - c) Network Attached Storage
 - d) Network Attacked Storage
28. What are types of optical disks
- a) CD-ROM, CD-R, CD-W, DVD-ROM, DVD-R, DVD-W
 - b) CD-ROM, CD-R, CD-RW, DVD-ROM, DVD-R, DVD-RW
 - c) CD, CD-R, CD-RW, DVD, DVD-R, DVD-RW
 - d) CD-RAM, CD-R, CD-RW, DVD-RAM, DVD-R, DVD-RW
29. What is the addressing mode of the instruction "LOAD R1, (R2)", where Ri is a CPU register
- a) Direct
 - b) Indirect
 - c) Immediate
 - d) Relative
30. Which is a correct name of SDRAM
- a) Statistical DRAM
 - b) Synchrone DRAM
 - c) Static DRAM
 - d) Synchronous DRAM
31. Which is NOT a drawback of cache look through architecture
- a) Hit is slow
 - b) Expensive
 - c) Miss is slow
 - d) Complicated design
32. SP or Stack Pointer is the register that always points to ... of the stack
- a) the bottom
 - b) next frame
 - c) the middle
 - d) the top
33. What is the meaning of the instruction "SUBSTRACT R1, R2, R3", where Ri is a CPU register
- a) $R3 \leftarrow R1 - R2$
 - b) $R1 \leftarrow R1 - R2 - R3$
 - c) $R1 \leftarrow R2 - R3$
 - d) $R2 \leftarrow R1 - R3$

34. The patterns of ... represent the information stored in optical disks
- a) holes and lands
 - b) pits and hills
 - c) holes and hills
 - d) pits and lands
35. When a program is executed, the operating system loads the program code stored in ... memory into ... memory
- a) internal / external
 - b) disk / RAM
 - c) external / internal
 - d) ROM / RAM
36. A physical HDD can be divided into several parts for better use and management. Each part is called...
- a) a partition
 - b) a section
 - c) an area
 - d) a logical drive
37. What are the functions of computer keyboards
- a) Input data and Control
 - b) Output data and Control
 - c) Input data and Storage
 - d) Import data and Storage
38. Internal memory is used to store instructions and data for CPU to process. Internal memory consists of ...
- a) ROM and SAM
 - b) ROM and RAM
 - c) ROM and SAM
 - d) ROM and RAM
39. Common arithmetic instructions include ...
- a) ADD, SUB, MULTIPLY, DIVIDE
 - b) ADD, SUBTRACT, MULTIPLY, DIVIDE
 - c) ADC, SUBTRACT, MULTIPLY, DIVIDE
 - d) ADD, SUB, MUL, DIV

40. What are the characteristics of RAM memory
- a) Volatile memory, semiconductor memory, random access memory
 - b) Volatile memory, magnetic memory, read only memory
 - c) Non-volatile memory, semiconductor memory, random access memory
 - d) Non-volatile memory, magnetic memory, random access memory
41. What is the meaning of the instructions "LOAD 1000, (R1)", where R1 is a CPU register
- a) $M[1000] \rightarrow M[R1]$
 - b) $1000 \rightarrow R1$
 - c) $M[1000] \leftarrow R1$
 - d) $M[1000] \leftarrow M[R1]$
42. What are the elements of cache direct mapping address
- a) Tag, Line, Byte
 - b) Tag, Line, Word
 - c) Tague, Lane, Word
 - d) Tab, Line, Word
43. What are advantages of SSDs over HDDs in computers
- a) Higher accessing speed, Less power consumption, More compact in size
 - b) Bigger accessing speed, Less wire consumption, Less compact in size
 - c) Higher accessing speed, Less power consumption, Less compact in size
 - d) Higher accessing, Less expensive, More compact in size
44. What is the CD-ROM component that converts the reflective beam to output electric signal
- a) Rotation mirror
 - b) Beam splitter
 - c) Beam sensor
 - d) Laser diode
45. After CPU executes one instruction, the ... is loaded into PC register
- a) memory address
 - b) memory address of next instruction
 - c) next instruction of memory address
 - d) next instruction

46. What are cache miss reduction methods

- | | |
|--|--|
| a) Increase cache size, Increase cache associative level | b) Increase cache block size, Increase cache associative level |
| c) Increase cache size, Increase cache level | d) Increase cache block size, Increase cache level |

47. What are correct typical addressing modes

- | | |
|---|---------------------------------------|
| a) Immediate, Direct, Indexed, Relative | b) Immediate, Direct, Index, Relative |
| c) Immediate, Direct, Index, Relative | d) Immediate, Direct, Indexed, Memory |

48. What are common computer system buses

- | | |
|------------------------------|-------------------------------|
| a) ISA, EISA, PCI, AGP, PCIe | b) ISA, EISS, PCId, AGP, PCIe |
| c) ISA, TISA, PCI, PGP, PCIe | d) ISS, EISA, PCI, AGP, PClex |

49. Liquid crystals in LCD are semi-solid substances that are sensitive to...

- | | |
|--------------------------------|--------------------------------|
| a) temperature and electricity | b) temperature or electricity |
| c) heat and electricity | d) temperature and electronics |

50. What are logical elements of HDDs of computer systems

- | | |
|--|--|
| a) Partitions, sectors, files, directories | b) Partition table, sectors, files, root directory |
| c) Partition table, boot sector, file system, root directory | d) Partition, root sector, file system, boot directory |

51. What are 2 LCD types based on the light sources

- | | |
|-----------------------------|--------------------------|
| a) Backlight and Reflective | b) Backlit and Reflect |
| c) Backlit and Reflective | d) Backlight and Reflect |

52. An electron gun in CRT monitors is the element that produces ...

- | | |
|---------------------|-----------------------|
| a) electrons | b) light beam |
| c) an electron beam | d) an electronic beam |

53. What are bit widths of ISA-AT and EISA buses
- a) 8 and 16
 - b) 16 and 24
 - c) 24 and 32
 - d) 16 and 32
54. Cache works based on 2 basic principles, including...
- a) space locality and time locality.
 - b) space locality and temporal locality.
 - c) spatial locality and temporal locality.
 - d) spatial locality and time locality.
55. Addressing modes are methods that ... are organized.
- a) command operands
 - b) instruction parts
 - c) instruction operands
 - d) instruction elements
56. What are NOT elements of an instruction execution cycle
- a) Memory read cycle, Memory write cycle, Peripheral read cycle, Interrupt acceptance cycle
 - b) Instruction fetch cycle, Memory read cycle, Peripheral read cycle, Bus free cycle
 - c) Instruction fetch cycle, Memory read cycle, Peripheral read cycle, Bus read cycle
 - d) Memory read cycle, Memory write cycle, Peripheral read cycle, Bus free cycle
57. Given two instructions: "ADD R1, R1, R3" and "SUB R4, R1, R2", where Ri is CPU register, what is the CPU pipeline conflict with these instructions
- a) Resource conflict
 - b) Data conflict
 - c) No conflict
 - d) Branching conflict
58. What are common interconnect interfaces of SSDs to computer systems?
- a) SAS, SATA, M.2, U.8, PCIe
 - b) SAS, SATA, M.2, U.2, PCIe
 - c) SAS, PATA, M.6, U.8, PCIe
 - d) ATA, SATA, M.2, U.2, PCI

59. The future trend of cache is it getting bigger and bigger to better support for multi-tasking...
- a) multi-core and multi-processor systems.
 - b) multi-cache and multi-core systems.
 - c) parallel processing and grid computing.
 - d) parallel processing and multi-core CPU systems.
60. Instruction execution is divided into phases or stages. One instruction can be executed in 5 stages, including...
- a) IC, ID, EX, MEM, WR
 - b) IS, ID, EX, MEN, WB
 - c) IF, ID, EX, MEM, WB
 - d) IF, IR, EX, MEM, WR
61. If memory only supports ... read/write operation at a time, ... memory accesses from CPU pipeline will cause resource conflicts.
- a) two / multiple
 - b) one / simultaneous
 - c) one / multiple
 - d) two / simultaneous
62. Two major types of RAM memories are...
- a) SRAM, DRAM.
 - b) SRAM, SDRAM.
 - c) RAM, DDR RAM.
 - d) SAM, DRAM.
63. What are characteristics of cache memory?
- a) very fast, small in size, high cost per bit
 - b) very fast, large in size, high cost per bit
 - c) fast, small in size, low cost per bit
 - d) very fast, small in size, very high cost per bit
64. What are the solutions for conditional branching conflicts of the CPU pipeline?
- a) Cancelled branching, Branch cancellation
 - b) Cancelled branching, Branch prediction
 - c) Delayed branching, Branch prediction
 - d) Delayed branching, Branch cancellation
65. SRAM is faster than DRAM because its bits have the ... and SRAM chips accept all address bits at a time.
- a) simple structure
 - b) symmetric structure
 - c) complex architecture
 - d) symmetric architecture

66. The Control Unit (CU) decodes the instruction and generates ... for other units, such as ALU to execute the instruction.
- a) command
 - b) request
 - c) execution signal
 - d) pulse
67. When a program code is loaded into internal memory, the address of the first instruction of the program is set onto ... register.
- a) FR
 - b) IR
 - c) MAR
 - d) PC
68. In computer functional block diagram, there are CPU, system bus,...
- a) internal memory and peripheral devices.
 - b) external memory and peripheral devices.
 - c) external memory and input devices.
 - d) internal memory and output devices.
69. Which is NOT a correct statement about CPU instructions?
- a) One instruction has its own function.
 - b) Instructions are stored in memory.
 - c) Instructions are saved from CPU to memory after execution.
 - d) Instructions are fetched from memory to CPU for execution.
70. SRAM is fast, but expensive, so it is usually used for small-in-size and fast memory elements, such as...
- a) registers, cache.
 - b) ROM, RAM.
 - c) cache, ROM.
 - d) cache, RAM.
71. What is an input device?
- a) Disk drive
 - b) Plotter
 - c) Printer
 - d) Monitor
72. What is an output peripheral device for computers?
- a) Mouse
 - b) Keyboard
 - c) Scanner
 - d) CD-R drive

73. CPU pipeline issues include...
- a) resource conflicts, data conflicts and branching instructions.
 - b) resource conflicts, data conflicts and looping instructions.
 - c) resource conflicts, memory conflicts and branching instructions.
 - d) memory conflicts, data conflicts and branching instructions.
74. Computer architecture is the science and art of selecting and interconnecting hardware components to create computers that meet ... goals.
- a) functions, performance and cost
 - b) functionality, performance and expensive
 - c) functionality, performance and cost
 - d) functionality, efficiency and cost
75. When a key is pressed, the keyboard controller detects the event and generate a corresponding... code.
- a) scan
 - b) char
 - c) UTF-8
 - d) ASCII
76. What are the characteristics of the cache set associative mapping technique?
- a) Simple, fast and fixed mapping
 - b) Simple, slow and flexible mapping
 - c) Complicated, fast and flexible mapping
 - d) Complicated, slow and flexible mapping
77. What is correct statement about MAR and MBR registers?
- a) MAR is slower than MBR.
 - b) MAR and MBR are interfaces between the CPU and system bus.
 - c) MAR is the input and MBR is the output of the CPU.
 - d) MAR is faster than MBR.
78. The numbers of minimum required HDDs of RAID 0, RAID 1 and RAID 10 are...
- a) 2, 2 and 6.
 - b) 2, 2 and 4.
 - c) 2, 4 and 4.
 - d) 4, 2 and 6.

79. What are components of PCI bus in PCI block diagram?
- a) ABH master, ABH slave, PCI referee, PCI interconnect
 - b) ABH server, ABH client, PCI referee, PCI interconnect
 - c) ABH server, ABH client, PCI arbiter, PCI interface
 - d) ABH master, ABH slave, PCI arbiter, PCI interface
80. What are the common sizes of CPU registers?
- a) 32, 64, 128, 256
 - b) 4, 8, 16, 32
 - c) 16, 32, 64, 128
 - d) 8, 16, 32, 64
81. What are the characteristics of the cache fully associative mapping technique?
- a) Complicated, slow and flexible mapping
 - b) Complicated, fast and flexible mapping
 - c) Simple, fast and fixed mapping
 - d) Simple, slow and flexible mapping
82. Generally, an instruction has two parts of opcode and...
- a) sources of operands.
 - b) values of operands.
 - c) addresses of operands.
 - d) destinations of operands.
83. Each bit of SRAM is based on ... while each bit of DRAM is based on...
- a) flip-flops / capacitors.
 - b) transitors / capacitors.
 - c) a flip-flop / a capacitor.
 - d) a transistor / a capacitor.
84. What is the correct name of AGP bus?
- a) Advanced Graph Port
 - b) Advanced Graphics Port
 - c) Accelerated Graphics Port
 - d) Accelerated Graph Port
85. Storage Area Network (SAN) is ... which is dedicated for storage services...
- a) a number of machines
 - b) a number of servers
 - c) a network of servers
 - d) a group of servers

86. A SRAM flip-flop usually uses ... transistors.
- a) 6, 8, 10
 - b) 4, 8, 12
 - c) 4, 6, 8
 - d) 8, 10, 12
87. What are the characteristics of magnetic disks?
- a) Volatile, hard storage, based on magnetic principles
 - b) Non-volatile, mass storage, based on magnetic principles
 - c) Volatile, mass storage, based on magnetic principles
 - d) Non-volatile, mass storage, based on ferromagnetic principles
88. Which is the name of the program stored in HDD boot sector?
- a) Rootstraper
 - b) Bootstrap loader
 - c) Rootstrap loader
 - d) Bootstraper
89. System buses are sets of wires to connect CPU with other components of a computer. System bus consist of...
- a) A bus, C bus and D bus
 - b) B bus, C bus and D bus
 - c) B bus, D bus and C bus
 - d) A bus, B bus and C bus
90. What is NOT a component of the instruction set architecture?
- a) The instruction set
 - b) Memory addressing
 - c) Address and data formats
 - d) Processor registers
91. What are the CPU registers?
- a) PC, IR, MBR, A, Y, Z
 - b) PC, RC, MCR, A, Y, Z
 - c) PK, IR, MRR, A, Y, Z
 - d) PC, IR, MBR, A, B, C
92. What type of pipeline conflicts is it, in which there are data dependency among consecutive instructions?
- a) Resource conflict
 - b) Data conflict
 - c) Branching conflict
 - d) No conflict

93. What are phases of a PCI transaction?

- a) Arbitration, Access, Data
- b) Arbitration, Address, Transfer
- c) Arbitration, Address, Data
- d) Arbiter, Address, Data

94. Common groups of CPU instructions include...

- a) data movement group, computational group, conditional and branching
- b) data processing group, computational group, conditional and branching group
- c) data movement group, arithmetic group, conditional and branching group
- d) data movement group, computational group, conditional and looping group

95. Which is the increment order of memory components in accessing speed?

- a) HDD, SSD, RAM, Registers, Cache
- b) HDD, SSD, RAM, Cache, Registers
- c) RAM, Cache, Registers, HDD, SSD
- d) SSD, HDD, RAM, Cache, Registers

96. What are common interconnect interfaces of HDDs to computer systems?

- a) PATA, SATA, SCSI, SAS
- b) PATA, KATA, SCSI, MAS
- c) PATA, SATA, SCSI, SAS, USB
- d) ATA, SATA, SCSI, SAS, USB

97. Cache is a memory element in the computer memory hierarchy and it acts as the ... between CPU and the main memory.

- a) storage
- b) memory
- c) immediate
- d) intermediary

98. What are types of cache misses?

- a) Compulsory misses, Capacity misses, Size misses
- b) Compulsory misses, Capacity misses, Conflict misses
- c) Compulsory misses, Storage misses, Conflict misses
- d) Compulsory misses, Capacity misses, Conflict misses

99. Why is DRAM usually cheaper than SRAM?
- a) Because DRAM uses less capacitors per memory cell than SRAM.
 - b) Because DRAM uses more transistors per memory cell than SRAM.
 - c) Because DRAM is simpler than SRAM.
 - d) Because DRAM uses less transistors per memory cell than SRAM
100. Size of CPU Accumulator (A) register is equivalent to CPU...
- a) word length.
 - b) data word length.
 - c) data word.
 - d) data size.
101. Instruction execution cycle is the ... that CPU completes the execution of the instruction.
- a) clock of time
 - b) period of time
 - c) periodic time
 - d) number of clocks
102. What are data transfer speeds of v1.x, v2.0, v3.0 PCIe 16 lanes?
- a) 4GB/s, 8GB/s, 12GB/s
 - b) 8GB/s, 16GB/s, 32GB/s
 - c) 8GB/s, 12GB/s, 16GB/s
 - d) 4GB/s, 8GB/s, 16GB/s
103. What are characteristics of CPU registers?
- a) Very fast, small in number and size
 - b) Very fast, small in counter and size
 - c) Very fast, small in size and low frequency
 - d) Fast, small and size
104. What is the correct name of TFT in LCD monitors?
- a) Thin Field Transistors
 - b) Thin Film Transistors
 - c) Thick Film Transistors
 - d) Thick Field Transistors
105. In CPU pipeline, the delayed branching technique is suitable for...
- a) very long pipelines.
 - b) medium pipelines.
 - c) long pipelines.
 - d) short pipelines.

106. What are types of printers?
- a) Typed-writer, Dot-matrix, Laser, Inkjet
 - b) Typed-writer, Pin-matrix, Laser, Inkjet
 - c) Typewriter-derived, Pin-matrix, Laser, Inkjet
 - d) Typewriter-derived, Dot-matrix, Laser, Inkjet
107. What is the data transfer speed of the third SATA interface generation?
- a) 6 Gb/s
 - b) 5 Gb/s
 - c) 8 Gb/s
 - d) 4 Gb/s
108. Which is NOT a correct cache replacement policies?
- a) Random
 - b) LRU
 - c) LIFO
 - d) FIFO
109. What is the elements that SSDs use to store data?
- a) solid-state flash memory chips
 - b) RAM memory chips
 - c) ROM memory chips
 - d) solid-state flesh memory chips
110. In cache spatial locality, if a memory location is accessed, the probability that its ... are accessed in ... is high.
- a) near neighbours / closed future
 - b) neighbours / future
 - c) closed neighbours / near future
 - d) closed neighbours / far future
111. What are the elements of cache set associative mapping address?
- a) Tab, Line, Word
 - b) Tag, Line, Word
 - c) Tag, Set, Byte
 - d) Tag, Set, Word
112. What are common SSD components?
- a) Flesh memory, controller, Cache RAM, M.2 interface
 - b) Flash memory, Controller, Cache, SATA interface
 - c) Flash memory, Controller, Cache, Connection interface
 - d) Flesh memory, Cache controller, Cache RAM, Connection interface

113. Control Unit (CU) gets instruction from ... register, decodes and generates signals to control other components of the system.

- a) IR b) MAR
c) PC d) MBR

119. What is the addressing mode of the instruction "LOAD Ri, X(PC)", where Ri is a CPU register?
- a) Direct
 - b) Relative
 - c) Indirect
 - d) Indexed
120. Computer memory components are usually classified based on...
- a) access types, information sustainability, capacity.
 - b) access speeds, information sustainability, manufacturing technologies.
 - c) access types, information capacity, manufacturing technologies.
 - d) access types, information sustainability, manufacturing technologies.
121. In look aside architecture, cache and memory are connected to ... and they "see" the CPU bus cycle...
- a) internal bus / at the same time.
 - b) system bus / at the same time.
 - c) system bus / at the different time.
 - d) internal bus / at the different time.
122. Why is cache is considered a smart memory?
- a) Because cache is a very fast memory.
 - b) Because cache can reduce the memory access time.
 - c) Because cache can predict the need of instruction and data of CPU.
 - d) Because cache can load enough instruction and data for the CPU.
123. What are the correct 3 major components of a basic CPU?
- a) CU, ALU, Registers
 - b) CU, Cache, Registers
 - c) GPU, ALU, Registers
 - d) CU, ALU, Cache
124. What are two basic techniques used in RAID storage system?
- a) Disk striping, Disk mirroring
 - b) Disk strip, Disk mirror
 - c) Disk striping, Disk rotating
 - d) Disk straping, Disk mirroring

125. What is the number of words can be transferred per clock cycle of DDR2 SDRAM?

- a) 6 b) 8
c) 4 d) 2

132. Which is a true statement about the performance role of cache memory?
- a) Cache reduces the fast speed of CPU to meet the slow speed of main memory.
 - b) Cache balances the fast speed of CPU and the slow speed of main memory.
 - c) The average access time of the memory system is near RAM access time.
 - d) Cache increases the speed of main memory to meet the CPU speed.
133. What is the cache replacement policy that has the lowest hit rate?
- a) FIFO
 - b) LRU
 - c) Random
 - d) LIFO
134. SCSI and SAS interfaces are usually used in ... computers.
- a) server
 - b) laptop
 - c) client
 - d) desktop
135. How many cache levels are there in most modern CPUs?
- a) 4
 - b) 3
 - c) 1
 - d) 2
136. What are correct types of RAID storage systems?
- a) RAID 0, RAID 2, RAID 4, RAID 6
 - b) RAID 3, RAID 5, RAID 7, RAID 9
 - c) RAID 2, RAID 4, RAID 6, RAID 8
 - d) RAID 1, RAID 3, RAID 5, RAID 7
137. What are 4 general purpose registers of the Intel 8086 CPU?
- a) AX, BX, SI, DI
 - b) A, B, C, D
 - c) AX, BX, CX, DX
 - d) A, X, Y, Z
138. What is the correct full name of RAID storage system?
- a) Redundance Array of Independence Disks
 - b) Redundant Array of Independent Disks
 - c) Reused Array of Independent Disks
 - d) Redundant Array of Individual Disks

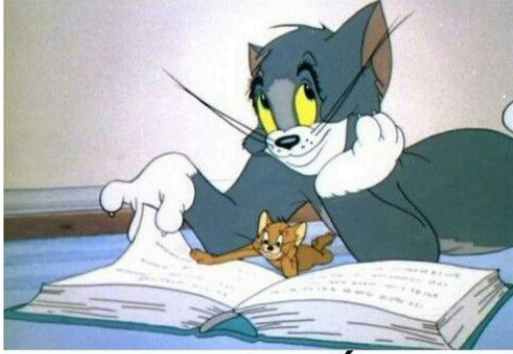
139. Keyboard is the ... device while monitor is the ... device of computers.
- a) input / output
 - b) standard output / standard input
 - c) standard input / standard output
 - d) internal / external
140. A solution for memory access conflicts in CPU pipeline is to ... to improve concurrent accesses.
- a) split L1 cache into I-Cache and D-Cache
 - b) add more RAM into the system
 - c) upgrade memory capacity of the system
 - d) split cache into multiple levels
141. The cache spatial locality usually applies to ... which have highly serial order in program space.
- a) data items or instructions
 - b) data items
 - c) instructions
 - d) data items and instructions
142. What are components participated in reading information from CD-ROMs?
- a) Laser diode, beam split, laser source, rotation mirror
 - b) Laser, splitter, sensor, mirror
 - c) Laser diode, beam splitter, laser sensor, rotation mirror
 - d) Laser diode, beam split, laser source, mirror rotation
143. What is the addressing mode of the instruction "LOAD R1, #1000", where R1 is a CPU register?
- a) Direct
 - b) Immediate
 - c) Indirect
 - d) Relative
144. What type of pipeline conflicts is it, in which a program have instructions, such as JUMP, CALL and RETURN?
- a) Branching conflict
 - b) No conflict
 - c) Data conflict
 - d) Resource conflict
145. Computer organization is a apart of computer architecture and also called...
- a) system design.
 - b) instruction set architecture.
 - c) macroarchitecture.
 - d) analysis and design.

146. What are functional keys on standard keyboards?
- a) F2, F4, F6, Fn, F8
 - b) F1, F3, F5, F9, Fn
 - c) F1, F2, F3, F4, F0
 - d) F1, F3, F5, F9, F11
147. What is the number of addresses of operands of the instruction "ADD R1, A", where R1 is a CPU register and A is a memory location?
- a) 1
 - b) 2.5
 - c) 2
 - d) 1.5
148. Control Unit (CU) uses clock to synchronize activities of functional units inside CPU and between ... components.
- a) CPU and GPU
 - b) CPU and internal
 - c) CPU and external
 - d) internal and external
149. Which is NOT a correct statement about instruction execution cycle and its components of traditional CPUs?
- a) One machine cycle consists of some clock cycles.
 - b) An execution stage may consist of some machine cycles.
 - c) An instruction execution cycle consists of some instruction execution stages.
 - d) An execution stage may consist of some execution phases.
150. What is the addressing mode of the instruction "LOAD Ri, X(Rind)", where Ri is a CPU register?
- a) Indexed
 - b) Indirect
 - c) Relative
 - d) Direct
151. System Design which includes all of the other hardware components within a computing system, such as...
- a) System connects and CPU off-load mechanisms.
 - b) System interconnects and CPU off-load mechanisms.
 - c) System connects and CPU on-load mechanisms.
 - d) System interconnects and CPU on-load mechanisms.

152. CRT monitor uses the ... emitted from its cathode to ... layers to create pictures.
- a) electric beam / phosphor
 - b) electronic beam / phosphor
 - c) electron beam / phosphor
 - d) electron beam / phosphor
153. What are the characteristics of the cache direct mapping technique?
- a) Simple, fast and fixed mapping
 - b) Complicated, slow and flexible mapping
 - c) Complicated, fast and flexible mapping
 - d) Simple, slow and flexible mapping
154. What is NOT the correct statement about ROM and RAM?
- a) ROM capacity is usually smaller than RAM capacity.
 - b) The information can only be read from ROM while the information can be read from and written.
 - c) ROM is only used to store instructions while RAM is only used to store instructions and data.
 - d) ROM is read only memory and RAM is random access memory.

155.

ÔN THI TƯỢNG TƯỢNG



THỰC TẾ



Nghĩ ngợi tý nhẻ

156. Vẽ vờ giải trí tý



157. Chúc cả lò nhà mình thi tốt ¹⁰⁰

158. CPU instruction is a ... which implements a single pre-defined operation of a processor.

- | | |
|----------------|-----------------|
| a) binary code | b) machine code |
| c) binary word | d) hexa word |

159. The number of addresses of operands of instructions may be 3,...

- | | |
|----------------------|--------------------|
| a) 2, 1, 1.5 and 0. | b) 2, 1, 0.5 and 0 |
| c) 2.5, 2, 1.5 and 1 | d) 2.5, 2, 1 and 0 |

160. What are the factors that affect cache performance?

- a) Cache size, cache separation, cache levels
- b) Cache size, cache split, multi-level cache
- c) Cache size, cache aggregate, multi-level cache
- d) Cache size, cache split, cache levels

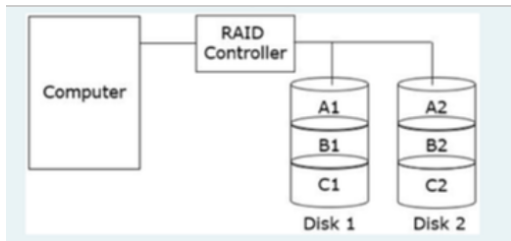
161. The computer input and output devices are also called ... devices.

- a) peripheral
- b) internal
- c) external
- d) periphered

162. What is the number of words can be transferred per clock cycle of DDR3 SDRAM?

- a) 8
- b) 6
- c) 4
- d) 2

163.



What is the RAID level in the following image?

- a) RAID 1
- b) RAID 0
- c) RAID 3
- d) RAID 2

164. Computer organization is the science that studies...

- a) computer structures and their using methodologies.
- b) computer components and their working methodologies.
- c) computer structures and their working methodologies.
- d) computer components and their running methodologies.

165. How many electron beams that the colour monitor uses to generate a pixel of a picture?

- a) 4
- b) 1
- c) 3
- d) 2

166. What are common logic instructions supported by ALU?

- a) NOT, NAND, OR, XOR
- b) NOT, AND, OR, XOR
- c) NOT, AND, NOR, XOR
- d) NOT, AND, OR, XORR

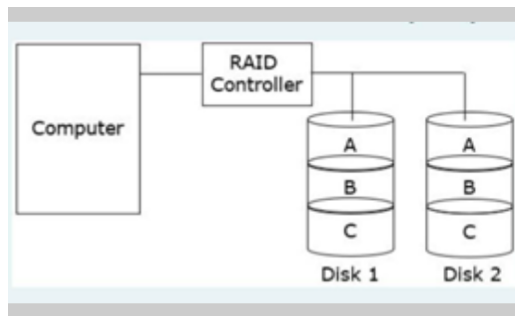
167. Why does DRAM memory need periodic refresh?

- a) Because electric charge in the capacitor of DRAM bits is leaking by nature.
- b) Because information stored in DRAM is removed automatically.
- c) Because electric charge stored in DRAM is removed automatically.
- d) Because electric charge in the transistor of DRAM bits is leaking by nature.

168. What is the addressing mode of the instruction "LOAD R1, (1000)", where R1 is a CPU register?

- a) Indirect
- b) Relative
- c) Direct
- d) Immediate

169.



What is the RAID level in the following image?

- a) RAID 2
- b) RAID 3
- c) RAID 1
- d) RAID 0

170. What is the technique used in RAID 0?

- a) Disk striping
- b) Disk striping and disk mirroring
- c) Disk mirroring
- d) Disk striping or disk mirroring

171. The components of memory system include...
- a) Registers, ROM, RAM, Cache, SSDs and HDDs.
 - b) Registers, ROM, RAM, Cache, External storages.
 - c) Registers, ROM, RAM, Cache, External HDDs.
 - d) Registers, ROM, REM, Cache, External Disks.
172. What is NOT a component of the instruction set architecture?
- a) Memory addressing modes
 - b) Address format
 - c) Processor registers
 - d) The instruction set
173. What is the meaning of the instruction "LOAD Ri, X(PC)", where Ri is a CPU register?
- a) $R_i \leftarrow M[X+PC]$
 - b) $R_i \leftarrow X + M[PC]$
 - c) $R_i \leftarrow X+PC$
 - d) $R_i \rightarrow M[X+PC]$
174. What are the status flags of the FR register?
- a) CF, DF, AF, ZF
 - b) TF, OF, AF, ZF
 - c) CF, OF, AF, ZF
 - d) CF, OF, AF, IF
175. One type of data conflict is Read After Write (RAW). A possible solution for RAW conflict is to...
- a) insert LO-OP instructions into the position between 2 instructions that have RAW.
 - b) insert NO-OP instructions into the position between 2 instructions that have RAW.
 - c) re-write the program.
 - d) re-order instructions of the program.
176. What is the addressing mode of the instruction "LOAD R1, 1000", where R1 is a CPU register?
- a) Indirect
 - b) Relative
 - c) Direct
 - d) Immediate
177. What is NO correct about internal bus?
- a) large bandwidth
 - b) communication channel of all bus components
 - c) support full duplex
 - d) faster than external buses

178. Which is true statement about the cost role of cache memory?
- a) If two system have the same performance, the system with cache is more expensive
 - b) If two system have the same performance, the system with cache is cheaper
 - c) If two system have the same cost, the system with cache is simpler
 - d) If two system have the same cost, the system without cache is faster
179. What is types of flags?
- a) status flags, control flags
 - b) status flags, control flags, data flags
 - c) status flags, control flags, address flags
 - d) status flags, control flags, condition flags
180. What is CPU includes?
- a) CU, ALU, Registers
 - b) CU, ALU, Cache
 - c) CU, ALU, Registers, Cache
 - d) CU, ALU, Memory
181. What is correct? "IR gets instruction from ... and forwards it to ... decoding"
- a) ROM, RAM
 - b) MAR, ALU
 - c) MBR, CU
 - d) PC, RAM
182. "MAR gets memory address of next instruction from ... and forwards it to ..."
- a) PC, address bus
 - b) IR, data bus
 - c) PC, control bus
 - d) IR, control bus
183. What is incorrect statement?
- a) MAR is the interface between the CPU and the address bus
 - b) MBR is the interface between the CPU and the data bus
 - c) MBR gets instruction from data bus and forwards it to IR
 - d) MAR gets memory address of next instruction from IR and forwards it to address bus
184. What is the correct full name of SSD in computers?
- a) Solid-State Driver
 - b) Solid-State Driver
 - c) Solid-Status Drive
 - d) Solid-State Drive

185. In computer systems, ... numbering system is used to represent data.
- a) Decimal
 - b) Hexadecimal
 - c) Binary
 - d) Octal
186. What is the function of computer mice?
- a) Input data
 - b) Control
 - c) Input data or Control
 - d) Input data and Control
187. Input and output (I/O) instructions are used to transfer data between ... and ...
- a) the CPU / external devices.
 - b) the computer / peripheral devices.
 - c) the CPU / peripheral devices.
 - d) the computer / internal devices.
188. What are common computer communication ports?
- a) PS/1, CON, LPT, USB, HDMI
 - b) PS/1, COM, LPTA, USB, HDMI
 - c) PS/2, COM, LPA, USB, HDMM
 - d) PS/2, COM, LPT, USB, HDMI
189. What are common arithmetic instructions supported by ALU?
- a) ADD, SBB, DIV, MUR, SHR, ROL
 - b) ADD, SUB, DIV, MUL, SHF, ROF
 - c) ADD, SUB, DIV, MUL, SHR, ROL
 - d) ADB, SUB, DIV, MUL, SHR, ROR
190. What are the inputs to the control unit (CU)?
- a) CLK, IR, ALU
 - b) CLK, PC, ALU flags
 - c) CLK, IR, flags
 - d) CLK, IR, ALU flags
191. What is the meaning of the instruction "ADD A, B, C", where A, B, C are memory locations?
- a) $M[A] \leftarrow M[A] + M[B] + M[C]$
 - b) $M[B] \leftarrow M[A] + M[C]$
 - c) $M[A] \leftarrow M[B] + M[C]$
 - d) $M[C] \leftarrow M[B] + M[A]$

192. Cache exchanges ... with CPU while it exchanges ... with main memory.
- a) individual data / blocked data
 - b) individual data items / data blocks
 - c) data blocks / individual data items
 - d) instruction items/instruction blocks
193. In CPU pipeline, the instruction execution is divided into stages and.... is executed by...
- a) each stage / a CPU component.
 - b) CPU components / stages.
 - c) a CPU component / each stage.
 - d) stages / CPU components.
194. What are major elements of cache memory?
- a) SDRAM, Cache controller, Tag Ram
 - b) SDRAM, Controller, Tag Ram
 - c) SRAM, Cache controller, Tag Ram
 - d) SRAM, Cache controller, Ram
195. What are the semiconductor memories?
- a) ROM, RAM, SSD
 - b) ROM, FDD, SSD
 - c) ROM, RAM, HDD
 - d) DVD, RAM, SSD
196. Which are correct names of file systems?
- a) FTA, NTFS, Ext2, Etx4
 - b) FAT, NTFS, Ext2, Ext4
 - c) FAT, NTFS, Ext2, etx4
 - d) FTA, TFS, Ext2, Ext4
197. In cache temporal locality, if a memory location is accessed, the probability that it is... is high.
- a) not accessed in near future
 - b) not accessed again in near future
 - c) accessed again in near future
 - d) accessed in near future
198. What are types of computer monitors?
- a) CRT, LCD, Retina
 - b) CRT, LCD, Amoled
 - c) CRT, LCD, OLED
 - d) CRT, LCD, Plasma

199. Hit is an event, in which CPU accesses a data item that is ... in cache and miss is an event, in which CPU accesses a data item that is ... in cache.
- a) not found / found
 - b) ready / not ready
 - c) available / not available
 - d) found / not found
200. Which is NOT a true memory type based on manufacturing technologies?
- a) Semiconductor memory
 - b) Magnetic memory
 - c) Optical memory
 - d) Volatile memory
201. What are main components of a lazer printer?
- a) Photosensitive drum, Laser light source, Rotating and laser modulator, Toner box
 - b) Photo drum, Laser light source, Rotating mirror and modulator, Toner cartridge
 - c) Photo drum, Laser light source, Rotating mirror and modulator, Toner box
 - d) Photosensitive drum, Laser light source, Rotating mirror and laser modulator, Toner cartridge
202. What are the types of ROM memory?
- a) PROM, EPROM, EEPROM, USB
 - b) ROM, EPROM, EEPROM, Flash memory
 - c) PROM, EPROM, ECPROM, Flash memory
 - d) PROM, EPROM, EEPROM, Flash memory
203. Which is the increment order of memory components in sizes?
- a) ROM, RAM, registers, cache, HDD
 - b) ROM, RAM, HDD, registers, cache
 - c) Registers, ROM, RAM, cache, HDD
 - d) Registers, Cache, ROM, RAM, HDD
204. What are the advantages of LCD over CRT monitors?
- a) Thinner and lighter, consume less power, larger size
 - b) Lighter, consume less power, larger viewable size
 - c) Thinner and lighter, consume less power, larger viewable size
 - d) Thinner, consume less power, larger viewable size

205. Common data movement instructions include ...
- a) MOV, LOD, STORE, PUSH.
 - b) MOVE, LOAD, STORE, PUSH.
 - c) MOVE, LOAD, STO, POP.
 - d) MOVE, LOAD, STORE, PUT.
206. Three components of computer architecture are ...
- a) Instruction set architecture, Microarchitecture, System Design.
 - b) Instruction set architecture, Macroarchitecture, System Design.
 - c) Instruction set architecture, Microarchitecture, Analysis and Design.
 - d) Instruction architecture, Microarchitecture, System Design.
207. Which is the cache level that is usually divided into 2 parts of I-cache and D-cache?
- a) Level 0
 - b) Level 1
 - c) Level 2
 - d) Level 3
208. What is the data transfer speed of the second SATA interface generation?
- a) 3 Gb/s
 - b) 4 Gb/s
 - c) 5 Gb/s
 - d) 6 Gb/s
209. What is the data transfer speed of the first SATA interface generation?
- a) 1.5 Gb/s
 - b) 2 Gb/s
 - c) 2.5 Gb/s
 - d) 3 Gb/s
210. A word is a group of ... bits
- a) 4
 - b) 8
 - c) 16
 - d) 32
211. Which of the following flags does not belong to Control flags?
- a) IF
 - b) TF
 - c) ZF
 - d) DF

212. Which of the following flags belongs to Status flags?
- a) DF
 - b) TF
 - c) IF
 - d) ZF
213. What are data transfer speeds of v1.x, v2.0, v3.0 PCIe 1 lane?
- a) 4GB/s, 8GB/s, 16GB/s
 - b) 4GB/s, 8GB/s, 12GB/s
 - c) 250MB/s, 500MB/s, 1GB/s
 - d) 250MB/s, 500MB/s, 750MB/s
214. What is the cache replacement policy that has the lowest miss rate?
- a) FIFO
 - b) LRU
 - c) Random
 - d) LIFO
215. In disk striping technique, data is divided into blocks and ... written into ... in parallel.
- a) each block is / a separate disk
 - b) they are / a separate disk
 - c) each block is / disks
 - d) they are / disks

Phím trả lời

- | | | |
|--|---|---|
| 1. b) a server / storage | 2. c) 4MHz, 6MHz, 8MHz | 3. c) Improve the system performance and reduce the manufacturing costs |
| 4. a) JUMP, CALL, RETURN, BRANCH-IF-CONDITION | 5. a) 4 | 6. b) CD-RW drive |
| 7. b) look aside and look through | 8. a) status of the result | 9. c) High performance, High reliability, Large volume |
| 10. c) 1 | 11. a) Cache usually resides inside the CPU | 12. b) stack |
| 13. b) Serial | 14. a) FDD, HDD, SSD, DVD, RAID, NAS, SAN | 15. d) Disk drive |
| 16. a) current executed | 17. c) Indirect mapping | 18. b) 1.5Gb/s |
| 19. b) two | 20. b) Peripheral Component Interconnect | 21. d) loop |
| 22. c) Disks, heads, tracks, sectors, cylinders | 23. a) Mechanical mouse, Optical mouse, Laser mouse | 24. a) Non-volatile memory, semiconductor memory, read only memory |
| 25. b) $R3 \leftarrow R2 + R1$ | 26. c) 700MB and 4.7GB | 27. c) Network Attached Storage |
| 28. b) CD-ROM, CD-R, CD-RW, DVD-ROM, DVD-R, DVD-RW | 29. b) Indirect | 30. d) Synchronous DRAM |
| 31. a) Hit is slow | 32. d) the top | 33. c) $R1 \leftarrow R2 - R3$ |
| 34. d) pits and lands | 35. b) disk / RAM | 36. a) a partition |
| 37. a) Input data and Control | 38. d) ROM and RAM | 39. b) ADD, SUBTRACT, MULTIPLY, DIVIDE |
| 40. a) Volatile memory, semiconductor memory, | 41. d) $M[1000] \leftarrow M[R1]$ | 42. b) Tag, Line, Word |

random access memory

- | | | |
|---|--|---|
| 43. a) Higher accessing speed, Less power consumption, More compact in size | 44. c) Beam sensor | 45. b) memory address of next instruction |
| 46. b) Increase cache block size, Increase cache associative level | 47. a) Immediate, Direct, Indexed, Relative | 48. a) ISA, EISA, PCI, AGP, PCIe |
| 49. a) temperature and electricity | 50. c) Partition table, boot sector, file system, root directory | 51. c) Backlit and Reflective |
| 52. c) an electron beam | 53. d) 16 and 32 | 54. c) spatial locality and temporal locality. |
| 55. c) instruction operands | 56. c) Instruction fetch cycle, Memory read cycle, Peripheral read cycle, Bus read cycle | 57. b) Data conflict |
| 58. b) SAS, SATA, M.2, U.2, PCIe | 59. d) parallel processing and multi-core CPU systems. | 60. c) IF, ID, EX, MEM, WB |
| 61. b) one / simultaneous | 62. a) SRAM, DRAM. | 63. a) very fast, small in size, high cost per bit |
| 64. c) Delayed branching, Branch prediction | 65. b) symmetric structure | 66. c) execution signal |
| 67. d) PC | 68. a) internal memory and peripheral devices. | 69. c) Instructions are saved from CPU to memory after execution. |
| 70. a) registers, cache. | 71. a) Disk drive | 72. d) CD-R drive |
| 73. a) resource conflicts, data conflicts and branching instructions. | 74. c) functionality, performance and cost | 75. a) scan |
| 76. c) Complicated, fast and flexible mapping | 77. b) MAR and MBR are interfaces between the | 78. b) 2, 2 and 4. |

CPU and system bus.

- | | | |
|--|--|--|
| 79. d) ABH master, ABH slave, PCI arbiter, PCI interface | 80. d) 8, 16, 32, 64 | 81. a) Complicated, slow and flexible mapping |
| 82. c) addresses of operands. | 83. c) a flip-flop / a capacitor. | 84. c) Accelerated Graphics Port |
| 85. c) a network of servers | 86. a) 6, 8, 10 | 87. b) Non-volatile, mass storage, based on magnetic principles |
| 88. b) Bootstrap loader | 89. a) A bus, C bus and D bus | 90. b) Memory addressing |
| 91. a) PC, IR, MBR, A, Y, Z | 92. b) Data conflict | 93. c) Arbitration, Address, Data |
| 94. a) data movement group, computational group, conditional and branching | 95. b) HDD, SSD, RAM, Cache, Registers | 96. a) PATA, SATA, SCSI, SAS |
| 97. d) intermediary | 98. b) Compulsory misses, Capacity misses, Conflict misses | 99. d) Because DRAM uses less transistors per memory cell than SRAM |
| 100. b) data word length. | 101. b) period of time | 102. d) 4GB/s, 8GB/s, 16GB/s |
| 103. a) Very fast, small in number and size | 104. b) Thin Film Transistors | 105. d) short pipelines. |
| 106. d) Typewriter-derived, Dot-matrix, Laser, Inkjet | 107. a) 6 Gb/s | 108. c) LIFO |
| 109. a) solid-state flash memory chips | 110. c) closed neighbours / near future | 111. d) Tag, Set, Word |
| 112. c) Flash memory, Controller, Cache, Connection interface | 113. a) IR | 114. b) Electrode charger for drum, Electrode charger for paper, Drying drum, Paper tray |
| 115. c) Because it can balance the speed of CPU and main memory better. | 116. b) system performance / manufacturing costs | 117. b) Magnetic memory |

- | | | |
|---|--|---|
| 118. a) Passive matrix and Active matrix | 119. b) Relative | 120. d) access types, information sustainability, manufacturing technologies. |
| 121. b) system bus / at the same time. | 122. c) Because cache can predict the need of instruction and data of CPU. | 123. a) CU, ALU, Registers |
| 124. a) Disk striping, Disk mirroring | 125. c) 4 | 126. c) a subsystem / components |
| 127. b) Disk mirroring | 128. b) CPU | 129. a) electronic imaging |
| 130. b) input and output | 131. c) Direct | 132. b) Cache balances the fast speed of CPU and the slow speed of main memory. |
| 133. c) Random | 134. a) server | 135. b) 3 |
| 136. a) RAID 0, RAID 2, RAID 4, RAID 6 | 137. c) AX, BX, CX, DX | 138. b) Redundant Array of Independent Disks |
| 139. c) standard input / standard output | 140. a) split L1 cache into I-Cache and D-Cache | 141. d) data items and instructions |
| 142. c) Laser diode, beam splitter, laser sensor, rotation mirror | 143. b) Immediate | 144. a) Branching conflict |
| 145. c) macroarchitecture. | 146. d) F1, F3, F5, F9, F11 | 147. d) 1.5 |
| 148. c) CPU and external | 149. d) An execution stage may consist of some execution phases. | 150. a) Indexed |
| 151. b) System interconnects and CPU off-load mechanisms. | 152. d) electron beam / phosphor | 153. a) Simple, fast and fixed mapping |

- | | | |
|---|---|--|
| 154. c) ROM is only used to store instructions while RAM is only used to store instructions and data. | 155. không có | 156. không có |
| 157. không có | 158. c) binary word | 159. a) 2, 1, 1.5 and 0. |
| 160. b) Cache size, cache split, multi-level cache | 161. a) peripheral | 162. a) 8 |
| 163. b) RAID 0 | 164. b) computer components and their working methodologies. | 165. c) 3 |
| 166. b) NOT, AND, OR, XOR | 167. a) Because electric charge in the capacitor of DRAM bits is leaking by nature. | 168. a) Indirect |
| 169. c) RAID 1 | 170. a) Disk striping | 171. b) Registers, ROM, RAM, Cache, External storages. |
| 172. b) Address format | 173. a) $R_i \leftarrow M[X+PC]$ | 174. c) CF, OF, AF, ZF |
| 175. b) insert NO-OP instructions into the position between 2 instructions that have RAW. | 176. c) Direct | 177. b) communication channel of all bus components |
| 178. b) If two system have the same performance, the system with cache is cheaper | 179. a) status flags, control flags | 180. a) CU, ALU, Registers |
| 181. c) MBR, CU | 182. a) PC, address bus | 183. d) MAR gets memory address of next instruction from IR and forwards it to address bus |
| 184. d) Solid-State Drive | 185. c) Binary | 186. b) Control |

- | | | |
|--|---|---|
| 187. b) the computer / peripheral devices. | 188. d) PS/2, COM, LPT, USB, HDMI | 189. c) ADD, SUB, DIV, MUL, SHR, ROL |
| 190. d) CLK, IR, ALU flags | 191. c) $M[A] \leftarrow M[B] + M[C]$ | 192. b) individual data items / data blocks |
| 193. a) each stage / a CPU component. | 194. c) SRAM, Cache controller, Tag Ram | 195. a) ROM, RAM, SSD |
| 196. b) FAT, NTFS, Ext2, Ext4 | 197. c) accessed again in near future | 198. d) CRT, LCD, Plasma |
| 199. d) found / not found | 200. d) Volatile memory | 201. d) Photosensitive drum, Laser light source, Rotating mirror and laser modulator, Toner cartridge |
| 202. d) PROM, EPROM, EEPROM, Flash memory | 203. d) Registers, Cache, ROM, RAM, HDD | 204. c) Thinner and lighter, consume less power, larger viewable size |
| 205. b) MOVE, LOAD, STORE, PUSH. | 206. a) Instruction set architecture, Microarchitecture, System Design. | 207. b) Level 1 |
| 208. a) 3 Gb/s | 209. a) 1.5 Gb/s | 210. c) 16 |
| 211. c) ZF | 212. d) ZF | 213. c) 250MB/s, 500MB/s, 1GB/s |
| 214. b) LRU | 215. a) each block is / a separate disk | |